# Md Ariful Islam Malik

mmalik3@crimson.ua.edu | Google Scholar | ORCID: 0009-0005-0885-5104 | DBLP | LinkedIn

## Summary

Computer Science PhD candidate at The University of Alabama (expected May 2027) with approximately six years of university-level faculty experience as a Lecturer and Assistant Professor, plus graduate teaching experience. My research interests include empirical software engineering, research software, peer code review, large language models (LLMs), and AI for software engineering (AI4SE). I have taught a broad range of computer science courses and supervised seven student project groups in areas including machine learning, neural networks, web systems, and Android-based systems.

## Research Interests

Empirical Software Engineering, Research Software, Peer Code Review, Large Language Models for Software Engineering, AI for Software Engineering (AI4SE)

## Education

2023 - 2027 (exp.) **Ph.D. in Computer Science** at The University of Alabama, Tuscaloosa, AL
Thesis: Improving Peer Code Review Practices in Research Software: A Study of Research Software Engineers
Advisor: Dr. Jeffrey C Carver

2021 – 2023 **M.S. in Computer Science** at Boise State University, Boise, ID
Project: An Automated Configuration Assistant for HydroFrame
Advisor: Dr. Catherine Olshanowsky

2010 - 2014 **B.S. in Computer Science and Engineering** at University of Dhaka, Bangladesh
Thesis: Similarity Computation for Collaborative Filtering Approaches Using Metric Aggregation
Advisor: Dr. Shabbir Ahmed

## Academic Experience

2023 – Current **Graduate Research Assistant** at The University of Alabama
I conduct empirical studies of code review in research software using surveys, interviews, and thematic analysis. I also investigate large language models for code review through an ongoing mixed-methods study involving repository mining, human and LLM-generated review comments, participant evaluation, and quantitative and qualitative analysis.

2022 – 2023 **Graduate Research Assistant** at Boise State University
I worked on the HydroFrame project, where I constructed a benchmarking suite using a variety of hydrological locations in the United States, developed an algorithm to recommend effective solver configurations, and developed a method to predict runtime and classify outcomes into three broad categories. I also contributed to the Full Waveform LiDAR project by writing test cases using GoogleTest (gtest).

2021 – 2022 **Graduate Teaching Assistant** at Boise State University
I tutored students in a Software Development in C programming course, graded the assignments, wrote test suites on GitHub for the assignments.

2019 – 2021 **Assistant Professor** at Bangladesh University of Business and Technology, Bangladesh
Continued teaching undergraduate computer science courses and supervising student thesis and project work.

2015 – 2019 **Lecturer** at Bangladesh University of Business and Technology, Bangladesh
Taught undergraduate courses including Software Design Pattern, Web Database Programming, Java, C++, C, Data Structures, Database Systems, and Neural Networks. Supervised student projects in areas including machine learning, neural networks, web systems, and Android-based systems.

## Research in Progress

2026 – 2027 (Exp)

- **AI-Powered Code Review Agents in Research Software**
  Investigating the types of coding issues identified by AI-powered code review agents in research software pull requests on GitHub, the extent to which developers act on AI review comments and how they address them, and the comment characteristics associated with developer action. A planned extension will examine factors developers report as influencing decisions to act or not act on AI review comments through interviews or surveys with original pull-request authors.

## Publications

- **Md Ariful Islam Malik**, Jeffrey C. Carver, and Nasir U. Eisty. **"Peer code review in research software development: The research software engineer perspective."** Empirical Software Engineering Journal (EMSE), vol. 31, no. 2, p. 42, 2026.
- **Md Ariful Islam Malik**, Jeffrey C. Carver, and Nasir U. Eisty. **"Peer code review in research software: Practices, improvements, and implementation challenges from interviews of RSEs."** In Proceedings of the Practice and Experience in Advanced Research Computing 2026: Resilient Roots + Empowered Communities (PEARC), pp. 1–8, 2026, Minneapolis, MN, USA.
- Md. Amzad Hossain, Istiaque Ahmed Suvo, Amitabh Ray, **Md Ariful Islam Malik**, and M. F. Mridha. **"Number Plate Recognition System for Vehicles using Machine Learning Approach."** In International Conference on Innovative Computing and Communication, Springer, January 2020, Singapore.
- Mahamudul Hasan, Shibbir Ahmed, **Md Ariful Islam Malik**, and Shabbir Ahmed. **"A comprehensive approach towards user-based collaborative filtering recommender system."** In International Workshop on Computational Intelligence (IWCI), pp. 159–164, IEEE, 2016, Dhaka, Bangladesh.

## Posters

- **Md Ariful Islam Malik**, Jeffrey C. Carver, and Nasir U. Eisty. **"Improving Peer Code Review in Research Software: From RSE Practices to an Ongoing Real-World Study of AI-Assisted Code Review."** US Research Software Engineer Conference (US-RSE), 2026, San Jose, CA, USA.
  https://doi.org/10.5281/zenodo.23467509

## Teaching Experience

2015 – 2021 ▪ **Undergraduate Teaching at BUBT**
Software Design Pattern, Web Database Programming, Java, C++, C, Data Structures, Database Systems, and Neural Networks.

2021 – 2022 ▪ **Graduate Teaching Assistant**
Software Development in C, Boise State University.

## Mentoring and Advising

2015 – 2021 ▪ **Undergraduate thesis and project supervision**
Supervised seven undergraduate student groups, including one thesis group and six project groups, in areas including machine learning, neural networks, web systems, and Android-based systems. The thesis work resulted in a peer-reviewed publication. *"Number Plate Recognition System for Vehicles using Machine Learning Approach."*

2015 – 2020 ▪ **Student Advisor** at BUBT, Bangladesh
Served as a student advisor for a section of 40 students.

## Awards and Honors

2026 Best Student Paper Award, PEARC 2026
"Peer Code Review in Research Software: Practices, Improvements, and Implementation Challenges from Interviews of RSEs."

2026 Graduate School Travel Grant, The University of Alabama — PEARC 2026

2026 Graduate School Travel Grant, The University of Alabama — US-RSE 2026

2011 4th Place, Intra-Department Programming Competition, University of Dhaka, Bangladesh

2004 2nd position in state, Government Scholarship, Grade 8 Countrywide Examination, Bangladesh
2001 1st position in state, Government Scholarship, Grade 5 Countrywide Examination, Bangladesh

## Professional Service

2027 **Reviewer**, International Conference on Artificial Intelligence, Computer, Data Sciences, and Applications (ACDSA 2027) — research papers
2026 **Sub-reviewer**, International Symposium on Empirical Software Engineering and Measurement (ESEM 2026) — technical paper
2026 **Reviewer**, Practice and Experience in Advanced Research Computing (PEARC 26) — BoF and panel submissions
2026 **Reviewer**, US Research Software Engineer Conference (US-RSE 26) — BoF submissions

## Research and Technical Skills

| | |
|---|---|
| Research methods | Survey design, semi-structured interviews, thematic analysis, mixed-methods research, empirical analysis, human-subject evaluation, repository mining |
| Machine learning / NLP | TensorFlow, PyTorch, Keras, Hugging Face Transformers, NLTK, Gensim, spaCy |
| Parallel / HPC | Pthreads, OpenMP, Open MPI, CUDA programming, Slurm jobs |
| Programming | Python, C, C++, Java, Bash |
| Software tools | Git, Make, GoogleTest (gtest), OpenGL |
| Languages | Bengali and English (written and verbal) |

## Leadership

2021-22 **Communication Coordinator** at Bangladeshi Student Association, Boise State University

## References

1. **Dr. Jeffrey Carver**
   Professor, Department of Computer Science, The University of Alabama, Tuscaloosa, AL
   Email: carver@cs.ua.edu
2. **Dr. Nasir Eisty**
   Assistant Professor, Department of EECS, University of Tennessee Knoxville, TN
   Email: neisty@utk.edu
3. **Dr. Muhammad Firoz Mridha**
   Professor, Department of Computer Science, American International University Bangladesh
   Email: firoz.mridha@aiub.edu